

Report
On
CISCO NETWORKING ACADEMY
Network Technician Career Path

Submitted to
School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh
In partial fulfilment of the requirement of the degree of
MCA



Submitted By
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Batch: 2025-2027

2026-27

CANDIDATE'S DECLARATION

I, Abhishek Yadav, hereby declare that the internship report titled "**Cisco Networking Academy: Network Technician Career Path**" has been completed by me to fulfil the requirement for the award of the degree of Master's of Computer Application in CSE. All the references have been quoted, and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Abhishek Yadav

Roll No.: 25SCS2040000921

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to Cisco Networking Academy and IILM University for providing this learning opportunity, and to my instructor for their guidance throughout the program.

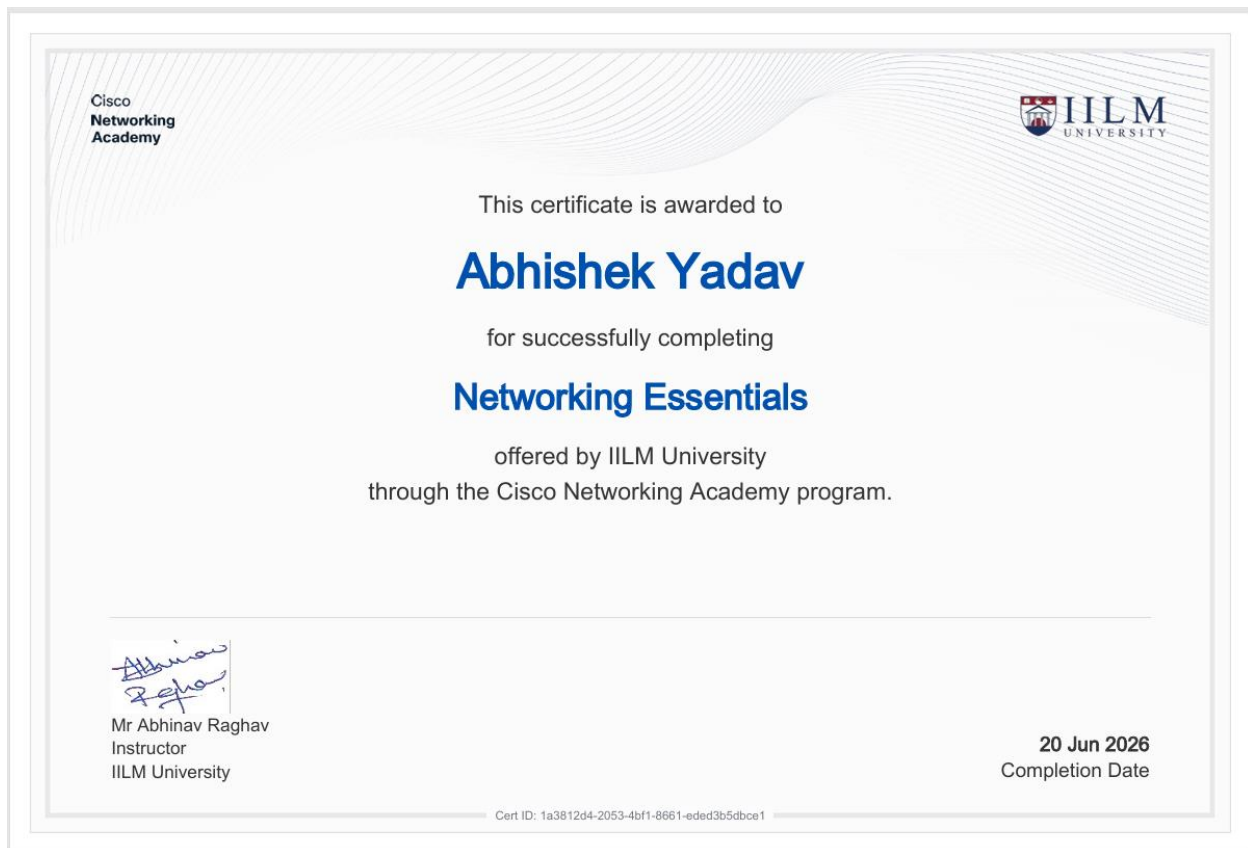
I express my gratitude to my parents for their blessings.

Signature : _____

Student Name : Abhishek Yadav

Roll No. : 25SCS2040000921

INTERNSHIP COMPLETION CERTIFICATES



Certificate title: Network Technician Career Path (Networking Essentials)

Certificate ID: 1a3812d4-2053-4bf1-8661-eded3b5dbce1

Date of issue: 20 Jun 2026

URL: https://www.credly.com/badges/6f13f487-d9e9-4347-864b-32a0ec4fec22/public_url

Cisco
Networking
Academy



This certificate is awarded to

Abhishek Yadav

for successfully completing

Getting Started with Cisco Packet Tracer

offered by IILM University
through the Cisco Networking Academy program.

A handwritten signature in blue ink, appearing to read "Abhinav Raghav".

Mr Abhinav Raghav
Instructor
IILM University

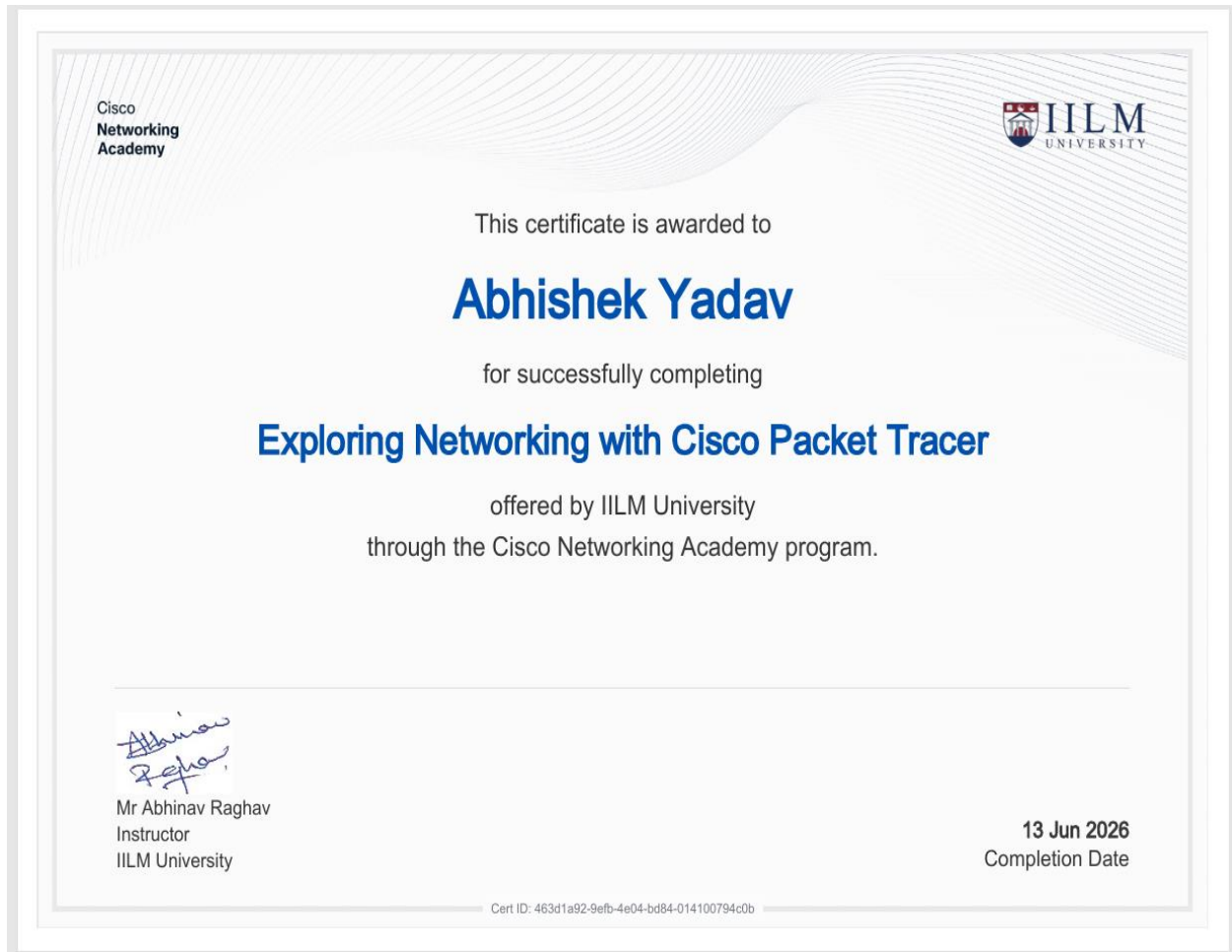
16 Jun 2026
Completion Date

Cert ID: ad8d47db-a7d2-4642-b442-7d52cd8a3be7

Certificate title: Getting Started with Cisco Packet Tracer

Certificate ID: ad8d47db-a7d2-4642-b442-7d52cd8a3be7

Date of issue: 16 Jun 2026



Certificate title: Exploring Networking with Cisco Packet Tracer

Certificate ID: 463d1a92-9efb-4e04-bd84-014100794c0b

Date of issue: 13 Jun 2026

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6. Executive Summary

This internship report documents the practical competencies acquired during the Cisco Networking Academy's Network Technician Career Path, completed through IILM University. The program was designed to take a learner from foundational networking theory through to applied, hands-on simulation, and it reflects the way real network technicians are expected to think: understand the underlying concept first, then prove that understanding by building, testing, and troubleshooting an actual network.

Over the course of the program, the learning path progressed steadily from core theoretical fundamentals — addressing, switching, and routing — to direct, tool-based practice inside Cisco Packet Tracer. Each module built on the one before it, so that by the time the final module was reached, the concepts of VLAN segmentation, inter-VLAN routing, and secure device access could be combined into a single, coherent design rather than treated as isolated topics. The training path integrated three modules:

- **Module 1: Networking Essentials**
- **Module 2: Getting Started with Cisco Packet Tracer**
- **Module 3: Exploring Networking with Cisco Packet Tracer**

Building on these modules, this report applies the acquired knowledge to a practical branch-office networking project, "SmartBranch 360", which involves designing, building, securing, and validating a small office network using Cisco Packet Tracer and a Python-based configuration checker. The project was chosen deliberately because it mirrors a task a junior network technician is genuinely likely to encounter: a business needs a new site brought online quickly, correctly, and securely, with a way to verify that the configuration matches the intended design before it is handed over.

Module 1: Networking Essentials

- **Core Concepts Covered:** Networking fundamentals, the OSI and TCP/IP models, IP addressing and subnetting, switching and routing basics, and how these layers work together to move data reliably across a network.
- **Technical Mechanisms:** Studied Ethernet LAN operation, VLAN segmentation, DHCP and DNS services, and basic routing between networks, with an emphasis on why each mechanism exists and what problem it solves.
- **Practical Exposure:** Configured end devices, verified connectivity using ping/traceroute concepts, and explored network addressing schemes, building the habit of verifying a design rather than assuming it is correct.

Module 2: Getting Started with Cisco Packet Tracer

- Core Concepts Covered: Packet Tracer interface, device placement, cabling conventions, and the difference between real-time and simulation mode.
- Technical Mechanisms: Building simple topologies, connecting PCs, switches, and routers, and testing connectivity in real-time and simulation modes to see exactly how packets move step by step.
- Practical Exposure: Hands-on exercises creating and testing basic LAN topologies within Packet Tracer, which formed the working foundation later reused for the SmartBranch 360 build.

Module 3: Exploring Networking with Cisco Packet Tracer

- Core Concepts Covered: Multi-device topologies, VLAN trunking, inter-VLAN routing, wireless access points, and access control lists.
- Technical Mechanisms: Router sub-interfaces, DHCP configuration per VLAN, NAT for internet access, and SSH-based secure management, each configured and tested individually before being combined.
- Practical Exposure: Built and troubleshot multi-VLAN topologies, validated traffic isolation between segments, and practiced fault injection and diagnosis — directly rehearsing the workflow later used in the SmartBranch 360 project.

7. Project Description

7.1 Introduction

This project, titled "SmartBranch 360," applies the concepts learned during the Network Technician Career Path to a real-world scenario: designing and securing a small branch-office network. A company opening a new branch office need wired employee connectivity, guest Wi-Fi, an internal server, internet access, and secure management of network devices. Currently there is no working design and no simple way to verify that the network is configured correctly, which means every change carries a real risk of an outage that is difficult to trace back to its cause.

This project designs and builds the branch network in Cisco Packet Tracer, verifies that traffic flows correctly across VLANs, deliberately introduces faults to practice structured troubleshooting, and develops a lightweight Python assurance tool that validates the network design and configuration. The intent is not only to produce a working topology, but to produce one that can be explained, re-verified after every change, and handed over with confidence — the same standard expected of a working network technician in the field.

7.2 Organization Profile

Cisco is a technology organization known for networking, cybersecurity, collaboration, and digital-learning initiatives. Cisco Networking Academy provides structured learning resources that help students and professionals build practical skills in networking, using tools such as Cisco Packet Tracer that closely mirror the behaviour of real enterprise-grade equipment.

The Network Technician Career Path, offered through IILM University via Cisco Networking Academy, gives learners the opportunity to study industry-relevant networking concepts through guided modules and hands-on simulation. For this report, the internship learning has been applied to a branch-office network design project. The project is presented as an academic implementation inspired by the concepts learned during the internship, structured to reflect how a small organization's IT team would actually plan and roll out a new site.

Organization details:

Field	Details
Program	Network Technician Career Path
Learning Platform	Cisco Networking Academy
Internship Mode	Online/Virtual
Student Name	Abhishek Yadav
Institute	IILM University, Greater Noida
Instructor	Mr. Abhinav Raghav

7.3 Problem Statement

A company is opening a new branch office. The branch requires wired employee connectivity, guest Wi-Fi, an internal server, internet access, and secure management of network devices. At present, there is no working network design and no straightforward way to verify whether the configuration is correct.

The task is to design and build this branch network in Cisco Packet Tracer, test that traffic flows correctly between segments, intentionally introduce faults to practice troubleshooting, and create a small Python assurance tool that validates the design and highlights likely configuration problems.

7.4 Project Objectives

Obj. No.	Objective	Related Activity
1	Design a segmented branch-office network	Plan VLANs, subnets, and device roles
2	Build the topology in Packet Tracer	Add router, switches, AP, server, and endpoints; configure cabling
3	Implement VLAN segmentation and routing	Configure trunking and router sub-interfaces for inter-VLAN routing
4	Provide automatic addressing	Configure DHCP per VLAN and verify DNS/internet reachability via NAT

Obj. No.	Objective	Related Activity
5	Enforce network security	Block guest access to server VLAN; restrict SSH to management VLAN
6	Practice structured troubleshooting	Inject and diagnose at least 5 fault scenarios
7	Automate configuration checking	Build a Python tool that validates the VLAN/IP plan and reports issues

7.5 Scope of the Project

The project covers the complete workflow from network planning to validation: topology design, VLAN and IP addressing, device configuration, security enforcement, fault injection and troubleshooting, and automated configuration checking using Python. Each stage was treated as a checkpoint — a stage was not considered complete until it could be verified, rather than assumed to work simply because the configuration had been typed in correctly.

The project does not attempt to replace a production-grade enterprise network deployment. It does not use real company infrastructure or live public IP addresses, and it does not claim that the Python checker replaces professional network-monitoring tools; it is a supporting aid that speeds up verification, not a substitute for hands-on testing or professional-grade monitoring software.

Included in Scope	Outside the Initial Scope
Simulated branch topology in Packet Tracer	Live/production network deployment
VLAN segmentation (Employee, Guest, Server, Management)	Multi-site WAN or cloud-based SD-WAN design
DHCP, DNS reachability, and NAT for internet access	Advanced enterprise security appliances (firewalls, IDS/IPS)
Access control for guest isolation and SSH management	Wireless enterprise authentication (802.1X/RADIUS)
Python-based configuration/plan checker	Fully automated network orchestration
Documented fault scenarios and fixes	Real customer or organizational data

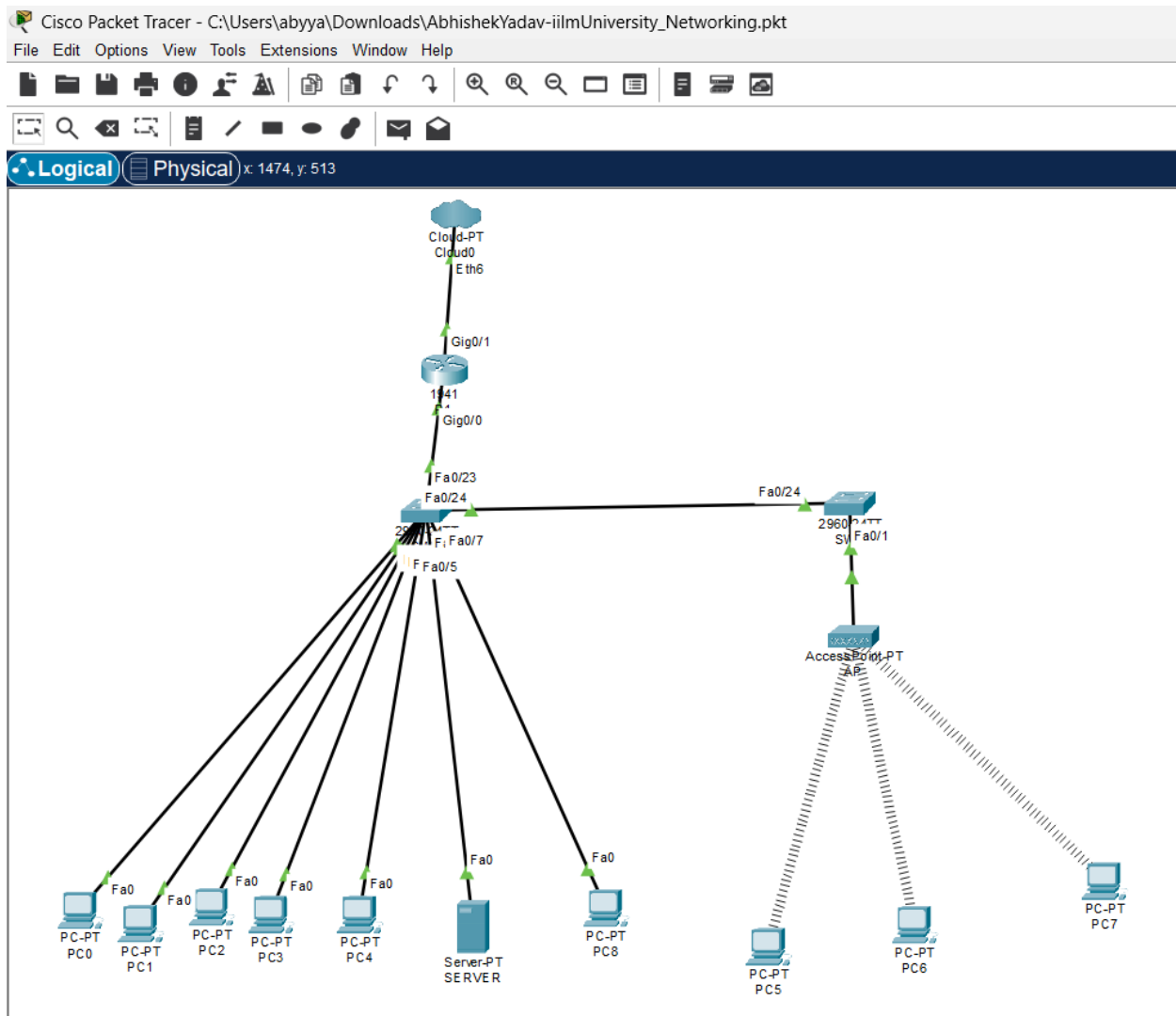
7.6 Technologies and Tools Used

Technology/Tool	Purpose
Cisco Packet Tracer	Building and simulating the branch-office network topology.
Router (IOS)	Inter-VLAN routing, NAT, and sub-interface configuration.
Layer 2/3 Switches	VLAN creation, trunking, and access-port assignment.
Wireless Access Point	Guest Wi-Fi connectivity.
DHCP & DNS services	Automatic IP addressing and name resolution per VLAN.
Python 3	Building the configuration/plan validation tool.
YAML/CSV	Storing the VLAN and IP requirement plan read by the Python tool.
Git/GitHub	Version control and project documentation.

7.7 System Architecture

The SmartBranch 360 topology consists of one router, two switches, one wireless access point, one internal server, an internet/cloud connection, and eight or more endpoints. Devices are segmented into four VLANs — Employee, Guest, Server, and Management — each mapped to a separate IP subnet and routed through router sub-interfaces (router-on-a-stick design). This design was chosen over a flat network specifically because it keeps guest traffic, business-critical server traffic, and administrative access from ever mixing on the wire, which is the single biggest security improvement a small branch office can make at low cost.

Figure 7.1: SmartBranch 360 high-level network topology



The architecture is intentionally modular — VLANs, subnets, or the classifier logic in the Python tool can be extended without redesigning the full topology.

7.8 Methodology

7.8.1 Network Planning

The branch requires four logical segments: Employee, Guest, Server, and Management. Each segment is assigned its own VLAN ID and IP subnet so that traffic can be isolated and routed with clear policy boundaries.

VLAN ID	VLAN Name	Subnet	Purpose
10	Employee	10.10.10.0/24	Wired employee workstations
20	Guest	10.10.20.0/24	Guest Wi-Fi via wireless AP
30	Server	10.10.30.0/24	Internal server(s)
99	Management	10.10.99.0/24	Device management (SSH access)

Table 7.4: VLAN and IP addressing plan

7.8.2 Topology Build in Packet Tracer

The topology is built using 1 router, 2 switches, 1 wireless access point, 1 server, an internet/cloud representation, and 8 or more endpoints. Access ports are assigned to the appropriate VLAN, and trunk links between switches and the router carry all four VLANs. Router sub-interfaces (e.g., G0/0.10, G0/0.20, G0/0.30, G0/0.99) provide inter-VLAN routing, each configured with the corresponding gateway IP and 802.1Q encapsulation.

7.8.3 Services Configuration

DHCP pools are configured per VLAN so that employee, guest, and management devices receive addressing automatically, with the server assigned a static IP. DNS reachability is verified for all internal VLANs, and NAT (overload/PAT) is configured on the router's internet-facing interface so that internal hosts can reach the internet using the ISP-assigned address.

7.8.4 Security Configuration

Policy	Implementation
Guest isolation	Access control list denies Guest VLAN (20) traffic to the Server VLAN (30) while permitting internet-bound traffic.
Management-only SSH	SSH access to routers/switches is permitted only from the Management VLAN (99); all other SSH attempts are denied.
Inter-VLAN control	Only the traffic explicitly required by the design (e.g., Employee → Server, Employee/Guest → Internet) is permitted.

Table 7.5: Security policy summary

7.8.5 Testing Normal Operation

After configuration, the following checks confirm the network behaves as designed:

- Employee PCs obtain a DHCP address, reach the internal server, and reach the internet.
- Guest devices reach the internet only — access to the Server and Management VLANs is blocked.
- Only the Management-VLAN host can establish an SSH session to network devices.

7.8.6 Fault Injection and Troubleshooting

At least five faults are deliberately introduced, one at a time, to practice structured diagnosis using symptom → root cause → fix reasoning:

#	Fault Scenario	Symptom	Root Cause	Fix
1	Missing VLAN on trunk	Guest PC has no connectivity	VLAN 20 not allowed on trunk SW1–SW2	Add VLAN 20 to the trunk's allowed list
2	Wrong default gateway	PC cannot reach other subnets	Gateway IP misconfigured on host/DHCP pool	Correct the gateway to match the VLAN sub-interface IP
3	DHCP pool exhausted/misconfigured	PC fails to obtain IP address	Wrong network/mask in DHCP pool	Correct DHCP pool network and default-router settings
4	ACL blocking DNS	Hosts can ping IPs but not resolve names	ACL denies UDP/53 traffic	Add a permit rule for DNS in the ACL
5	NAT misconfiguration	Internal hosts cannot reach internet	Interface not marked NAT inside/outside	Correctly mark interfaces and verify NAT pool/overload config

Table 7.6: Fault scenarios (fault cards)

7.8.7 Python Assurance Tool

A Python script reads the VLAN/IP requirement plan (YAML) and, optionally, device show-command output, then reports likely configuration problems in plain language — for example, a VLAN missing from a trunk, a mismatched gateway, or an ACL blocking required traffic. The tool is intended to speed up verification after each change rather than replace hands-on testing.

Figure 7.2: Project workflow

Define Requirements → Plan VLANs/IPs → Build Topology in Packet Tracer → Configure Services & Security → Test Normal Operation → Inject Faults → Run Python Checker → Diagnose & Fix → Re-verify → Document Results

7.8.8 Demo and Documentation

The working network is demonstrated end-to-end: normal operation is shown, one fault is injected, the Python tool is run to surface the likely cause, the issue is fixed in Packet Tracer, and connectivity is re-verified using ping/traceroute. Screenshots and outputs are recorded for the final report.

7.9 Expected Outcomes

The expected outcome is a working simulated branch-office network with correctly segmented VLANs, functioning DHCP/DNS/NAT services, enforced security policies for guest isolation and management access, and a reusable Python tool for configuration validation. Beyond the working topology itself, the project is expected to demonstrate a repeatable process: plan, build, secure, test, break, diagnose, fix, and re-verify — a cycle that applies to any future site the same design pattern is extended to.

Deliverables include the Packet Tracer topology file (SmartBranch360.pkt) with saved device configurations, a design document (topology diagram, VLAN/IP table, security rules, troubleshooting notes), a YAML/spreadsheet requirement file, the Python validation script with sample output, at least five documented fault cards, and a short demo recording.

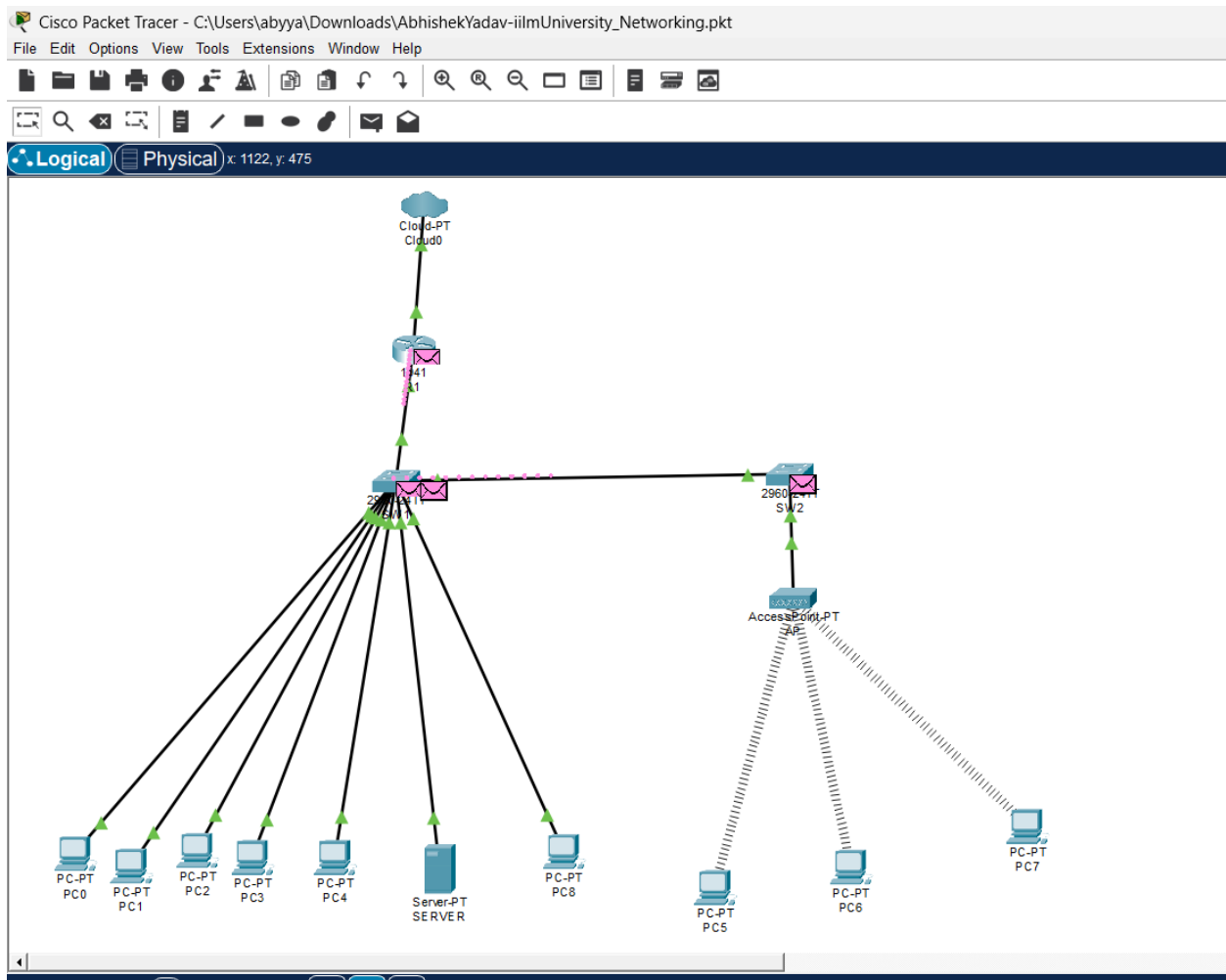
Recommended conclusion: The project successfully demonstrates how core networking concepts — VLAN segmentation, routing, DHCP/DNS/NAT services, and access control — can be combined into a secure, working branch-office design, validated with both manual testing and an automated Python checker. The exercise reinforced that a network is only as trustworthy as its verification process; building it correctly the first time matters less than being able to prove, quickly and repeatably, that it still works correctly after every subsequent change. The design is suitable for educational use and can be extended with enterprise security appliances, wireless authentication, and multi-site connectivity.

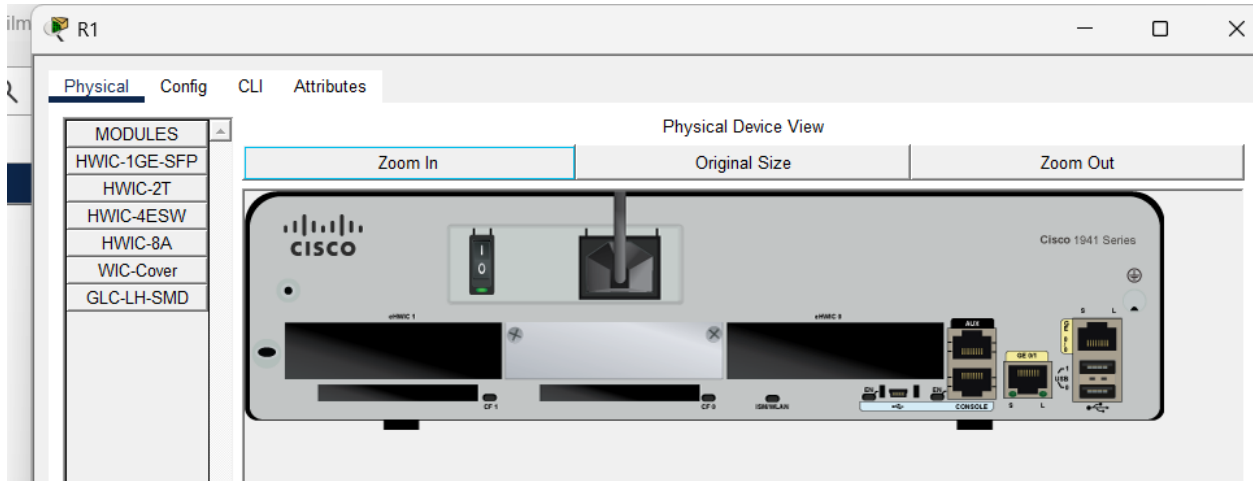
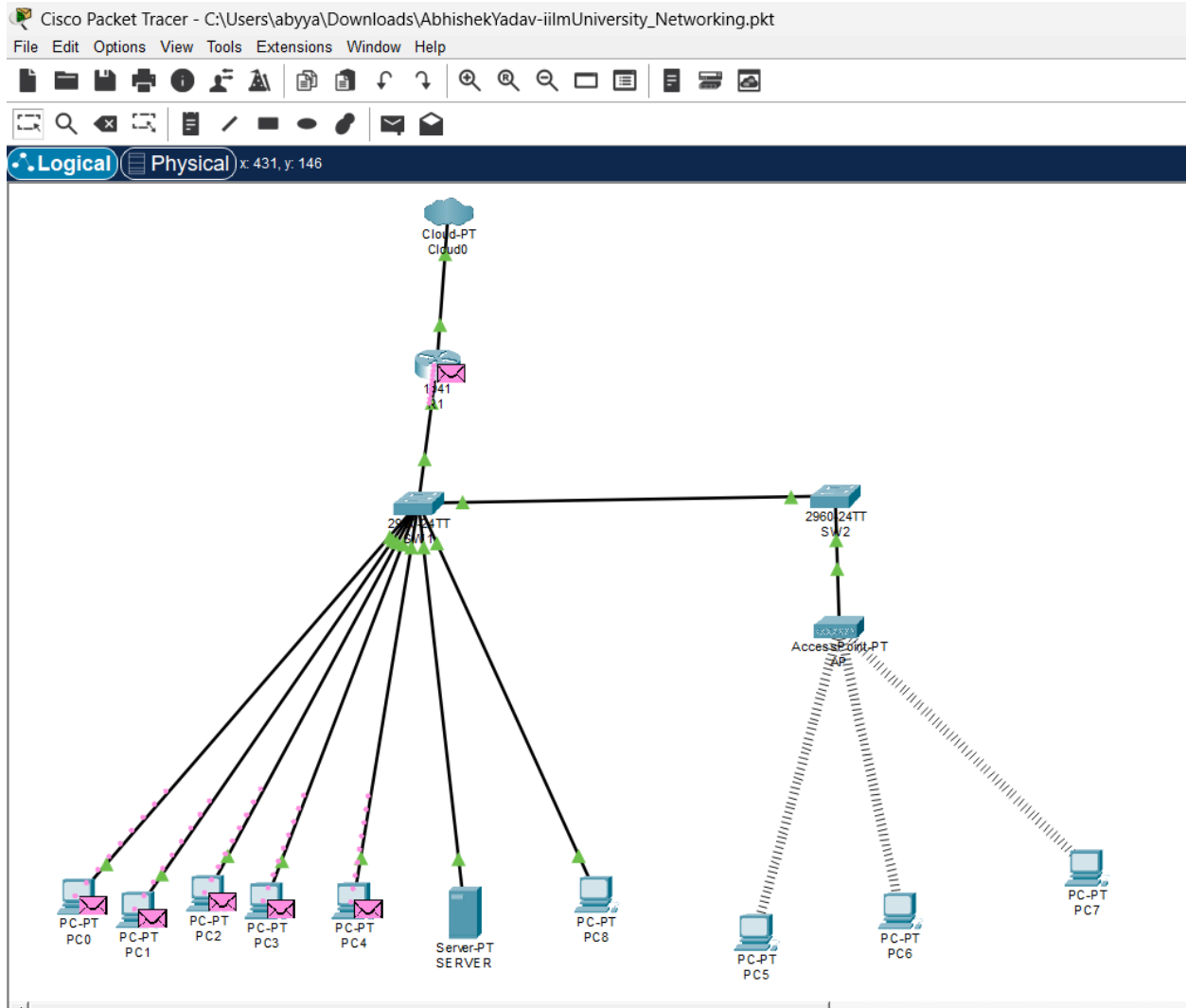
7.10 Certificates of Completion and Other Communication Proof

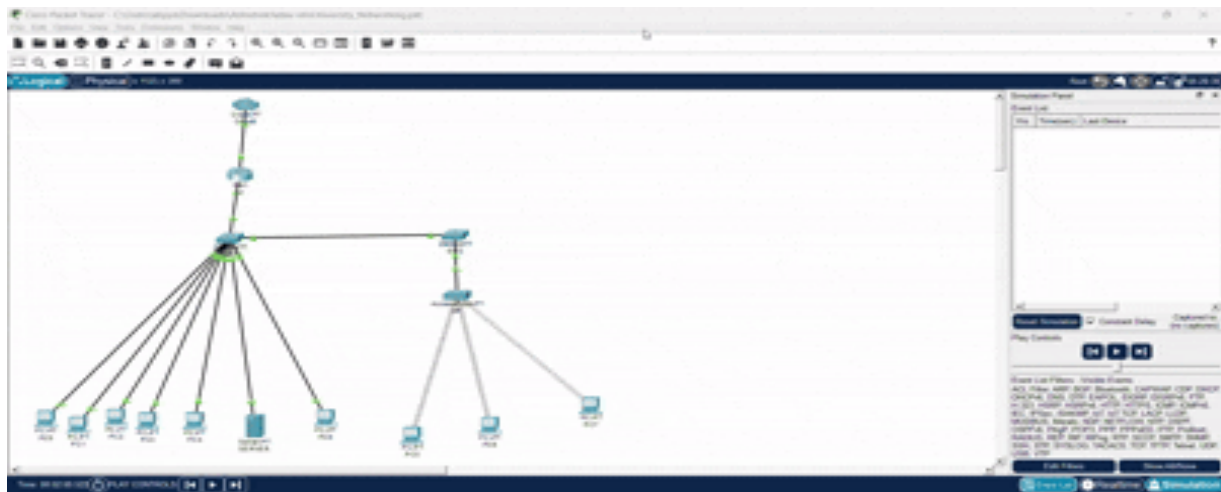
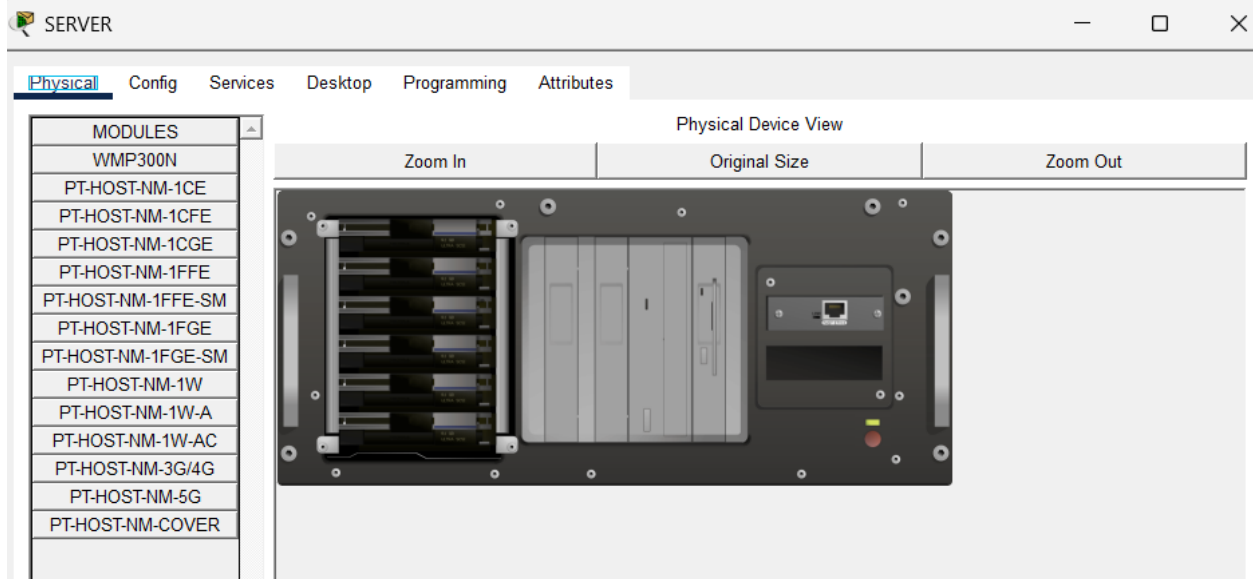
Attach the following documents after the project description or in the annexure, according to the college's instructions:

Annexure	Document to Attach
Annexure A	Cisco Networking Academy — Network Technician Career Path completion certificate.

Annexure	Document to Attach
Annexure B	Getting Started with Cisco Packet Tracer — course completion certificate.
Annexure C	Exploring Networking with Cisco Packet Tracer — course completion certificate.
Annexure D	SmartBranch360.pkt file and Python validation script/output.
Annexure E	Screenshots of the working topology, fault injection, and fix verification.







8. Bibliography/References

- [1] Cisco Networking Academy, "Networking Essentials Course," Cisco Networking Academy. Available at: <https://www.netacad.com/courses/networking-essentials>
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- [6] Python Software Foundation, "Python 3 Documentation," Python Documentation. Available at: <https://docs.python.org/3/>
- [7] PyYAML Developers, "PyYAML Documentation," Available at: <https://pyyaml.org/wiki/PyYAMLDocumentation>